A Prime-Free Interval and Factor Restrictions for Fortunate Numbers

A theorem note for OEIS A005235

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Abstract

For the Fortunate number $a(n)$, the smallest integer $m > 1$ for which the nth primorial plus m is prime, we prove an explicit interval of composite integers, two-sided bounds, a restriction on all prime factors of $a(n)$, and a sufficient condition for $a(n)$ to be prime. These are elementary facts, in part already stated in the OEIS; the open conjecture that *every* Fortunate number is prime is not established.

Definition and unresolved question

Let $p_1 = 2 < p_2 = 3 < \cdots$ be the primes and set

$$P_n = \prod_{j=1}^{n} p_j, \qquad a(n) = \min\{m > 1 : P_n + m \text{ is prime}\}, \qquad n \geq 1.$$

This is OEIS A005235, beginning $3, 5, 7, 13, 23, 17, 19, 23, 37, 61, \ldots$. Fortune conjectured that $a(n)$ is prime for all n. The following results do *not* settle this conjecture.

Theorem 1 (Prime-free interval and exact factor restriction)**.** *For every $n \geq 1$:*

1. *Every number $P_n + m$ with $2 \leq m < p_{n+1}$ is composite. Thus there is a block of $p_{n+1} - 2$ consecutive composite integers immediately following $P_n + 1$.*
2. *The Fortunate number is odd and satisfies*

$$\gcd(a(n), P_n) = 1, \qquad p_{n+1} \leq a(n) \leq P_n + 1.$$

3. *Every prime factor r of $a(n)$ satisfies $r \geq p_{n+1}$. Consequently, if $a(n)$ is composite, then*

$$a(n) \geq p_{n+1}^2.$$

Proof. For $2 \leq m < p_{n+1}$, choose any prime divisor r of m. Since there is no prime between p_n and p_{n+1}, one has $r \leq p_n$ and hence $r \mid P_n$. Therefore $r \mid (P_n + m)$. As $P_n + m > r$, this sum is composite; the number of such consecutive m is $p_{n+1} - 2$. This proves (1) and the lower bound in (2).

Write $Q = P_n + a(n)$; by definition Q is prime and $Q > P_n \geq 2$. If r were a common prime divisor of P_n and $a(n)$, then $r \mid Q$, contradicting $Q > r$. Thus the gcd is 1. Because P_n is even and Q is an odd prime, $a(n)$ is odd.

To prove the upper bound, apply Bertrand's postulate to the integer $P_n + 1 > 1$. There exists a prime Q' with

$$P_n + 1 < Q' < 2(P_n + 1).$$

Then $m' = Q' - P_n$ is an admissible integer exceeding 1 and obeys $m' < P_n + 2$. Hence $a(n) \leq m' \leq P_n + 1$.

Finally, if r is any prime factor of $a(n)$, the gcd restriction implies $r \nmid P_n$. Hence $r > p_n$, so $r \geq p_{n+1}$. A composite integer has at least two prime factors counting multiplicity, each at least p_{n+1}; this gives (3). □

Corollary 1 (Primality threshold and unboundedness)**.** *If $a(n) < p_{n+1}^2$, then $a(n)$ is prime. In addition $a(n) \to \infty$ as $n \to \infty$, and the sequence has infinitely many distinct values.*

Proof. The primality assertion follows from Theorem 1(3), since $a(n) > 1$. The inequality $a(n) \geq p_{n+1}$ and the infinitude of primes imply $a(n) \to \infty$. □

Small checks and status

n	P_n	p_{n+1}	$a(n)$	$P_n + a(n)$
1	2	3	3	5
2	6	5	5	11
3	30	7	7	37
4	210	11	13	223
5	2310	13	23	2333
6	30030	17	17	30047

The examples are illustrative; they are not the proof. The lower bound and the sufficient primality threshold are *explicitly noted* in A005235, with attribution to F. Firoozbakht and T. Ordowski, respectively. The interval interpretation and the elementary upper bound are presented here as a self-contained exposition. No novelty is asserted.

Limitations, references, and disclosure

Limitation. The theorem restricts the size and prime factors of hypothetical composite Fortunate numbers; it gives no unconditional proof that all $a(n)$ are prime. The upper bound $P_n + 1$ is deliberately elementary and far from sharp.

[1] OEIS Foundation, A005235: Fortunate numbers, definition, comments, and references (accessed November 13, 2026).

[2] A. Čejchan, M. Křížek, and L. Somer, "On Remarkable Properties of Primes Near Factorials and Primorials," *Journal of Integer Sequences* **25** (2022), Article 22.1.4 (cited in A005235).

[3] Bertrand's postulate (classical theorem on a prime between x and $2x$ for integer $x > 1$).

Authorship and verification. Primary author (AI): ChatGPT (OpenAI). The arguments were checked algebraically and against initial OEIS values; independent human peer review has *not* been performed.